

Atharva Rath

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Education

K.J Somaiya College of Engineering

B.Tech in Information Technology

Honours in Artificial Intelligence

Mumbai, Maharashtra

Aug 2023 – Present

Achievements: TechSprint by GDG KJSSE winner | 31st Rank IIT Ropar Predictathon (Round 1) | SIH 2025 External Round

Experience

Software Engineering Intern

Dec 2025

Trueview

Aurangabad, Maharashtra

- Designed and implemented a Retrieval-Augmented Generation (RAG) architecture, utilizing semantic search and vector embeddings to optimize the retrieval of complex product specifications and reduce internal query resolution time.
- Deployed a scalable AI backend integrated with a customized conversational interface, establishing secure API endpoints and robust data pipelines to ensure seamless user interaction and high-performance querying.
- Coordinated with internal manufacturing and operational teams to align Large Language Model (LLM) capabilities with strategic business workflows, delivering an automated knowledge-sharing platform that significantly boosted team productivity.

Software Engineering Intern

May 2025 – Jul 2025

Trueview

Aurangabad, Maharashtra

- Developed a fully functional e-commerce website for the subsidiary company, enhancing product visibility and driving a 20% increase in online sales within the first quarter.
- Engineered secure login systems, real-time notifications, and a scalable backend architecture using modern web technologies to ensure seamless user experience and performance.
- Collaborated cross-functionally with design, business, and product teams to align technical implementation with strategic business goals, resulting in an intuitive and conversion-optimized platform.

Projects

RAG-Powered Enterprise Knowledge System | Python, React, Vector Database, LLM APIs

- Architected a custom Retrieval-Augmented Generation (RAG) platform to streamline organizational access to complex hardware manufacturing specifications and product data.
- Implemented semantic search utilizing vector embeddings to parse unstructured technical documentation, delivering highly accurate, context-aware query responses.
- Deployed a secure, scalable conversational interface to automate knowledge retrieval workflows, significantly reducing internal data lookup times.

Automated AI Security & Vulnerability Scanner | Python, LLM APIs, Abstract Syntax Trees (AST), CLI

- Developed an automated Static Application Security Testing (SAST) pipeline that parses source code logic to identify potential vulnerabilities such as injection flaws and unauthorized data exposure.
- Integrated targeted Large Language Model (LLM) prompts to evaluate complex logic flaws, significantly reducing false positives and generating context-aware remediation code snippets.
- Packaged the analysis engine as a modular Command Line Interface (CLI) tool, enabling developers to execute rapid, pre-commit security audits directly from their local environment.

Fault-Tolerant Webhook Dispatch Engine | Go, Redis, Docker, Concurrency

- Architected an asynchronous event-delivery system in Go, utilizing worker pools and channels to concurrently process and dispatch thousands of high-volume webhook payloads.
- Engineered a reliable message-queuing architecture using Redis, implementing exponential backoff algorithms and automated retries to guarantee delivery against failing external endpoints.
- Designed a dead-letter queue (DLQ) mechanism to isolate and log permanently failed payloads, ensuring core system stability and preventing queue bottlenecks under heavy load.

Technical Skills

Languages & Frontend: Python, Go, Java, C++, JavaScript/TypeScript, React.js, Material-UI (MUI)

Backend & Databases: Node.js, Express.js, RESTful APIs, PostgreSQL, MongoDB, Redis, Vector DBs, Microservices

Cloud, AI & Tools - Docker, AWS, Git/GitHub, Linux, AI Agents, Computer Vision, RAG Architecture, LLMs, InfoSec